

Probing critical phases in quasiperiodic systems via subsystem information capacity

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We systematically investigate the entanglement dynamics of quasiperiodic systems across their extended, critical, and localized phases, aiming to identify dynamical signatures that can clearly distinguish the critical phase from the other two. Focusing on the extended Harper model, we complement the half-chain entanglement entropy with the spatially resolved subsystem information capacity (SIC) and demonstrate that the critical phase exhibits a pronounced spatial heterogeneity that is absent in the extended and localized phases. In the steady state, the SIC reveals a stepwise ramp as a function of subsystem size, reflecting an underlying fragmentation of the chain into weakly connected subregions. Dynamically, information initially localized within such a subregion can undergo coherent long-lived oscillations, dubbed *subregion echoes*, whose period scales with the subregion length, in quantitative agreement with a quasiparticle picture of confined quasiparticle reflections. We trace this internal fragmentation to the incommensurately distributed zeros (IDZs) in the off-diagonal hopping terms of the Hamiltonian. To establish the generality of the SIC as a diagnostic tool, we further apply it to a mobility-edge phase with coexisting extended and localized states and to a critical phase that does not originate from IDZ fragmentation, and show that the SIC can cleanly distinguish these scenarios through their distinct steady-state profiles, initial-site sensitivities and the presence of subregion echoes. Our results establish the SIC as a powerful real-space probe for diagnosing critical phases and for uncovering the bottlenecked connectivity that underlies their multifractal structure.

I. INTRODUCTION

Anderson localization, arising from destructive interference among multiple scattering paths in a disordered medium, is a cornerstone phenomenon in the study of quantum transport [1, 2]. Situated between perfectly periodic and fully random systems, quasiperiodic lattices provide a distinct and highly tunable setting for localization physics [3–14]. Unlike generic one-dimensional (1D) random potentials, quasiperiodic models can support not only fully extended and fully localized states, as illustrated by the prototypical quasiperiodic model known as the Aubry-André (AA) model [15], but also mobility edges that separate these two regimes within the same spectrum [5–9, 16–27]. Most intriguingly, they may host multifractal critical states and even critical phases in an intermediate regime between extension and localization [10–14, 28–41]. Such critical states are neither Bloch-like nor exponentially localized; instead, they are typically characterized by multifractal wave functions, local scale invariance, and anomalous transport dynamics, and are delocalized in both real and momentum space [28]. In generalized quasiperiodic models, such criticality can occupy a finite region of parameter space, forming a genuine critical phase rather than merely an isolated transition point [28–37, 39–41]. Recent theoretical advances have further sharpened the characterization of critical states and the associated phase boundaries, establishing a rigorous framework for identifying criticality and the mobility edges separating critical, extended, and

localized regimes [34, 41].

Despite these conceptual advances, the unambiguous identification of critical phases in realistic settings remains a nontrivial challenge. Conventional diagnostic tools, such as the fractal dimension extracted from eigenstate inverse participation ratios, provide a static, global characterization of the spectrum and offer limited insight into the dynamical transport properties that ultimately distinguish critical phases from their extended and localized counterparts. This motivates the search for complementary, real-space dynamical probes capable of directly sensing the distinctive spatial architecture of critical states [11, 13, 28]. Entanglement dynamics constitutes a promising candidate in this regard, as the growth and spreading of entanglement reflect the underlying structure of eigenstates and the transport properties of the system [42–50]. While the basic patterns of entanglement growth in extended and localized phases are well understood—rapid volume-law spreading in the extended phase [51] and strong suppression with area-law saturation in the localized phase [42, 46]—a systematic dynamical characterization of critical phases is far from complete. Recent studies have revealed that the entanglement entropy (EE) in critical phases exhibits a logarithmic scaling with subsystem size [52], yet how the multifractal nature of critical wave functions manifests in real-time entanglement dynamics, and how such dynamical signatures can be exploited to distinguish critical phases from extended and localized ones, remain largely unexplored. Motivated by this, the recently introduced subsystem information capacity (SIC) offers a particularly appealing real-space diagnostic [53, 54]. By tracking how quantum information initially encoded at a single site diffuses through the system, the SIC offers a spatially

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resolved, site-specific window into information flow, making it ideally suited to capture the heterogeneous connectivity that underpins critical phases.

In this work, we systematically investigate the entanglement dynamics of the extended Harper model across its extended, critical, and localized phases, with the goal of identifying dynamical signatures that can clearly distinguish these regimes. We complement the conventional half-chain EE with the SIC and demonstrate that, while the half-chain EE can separate localized from delocalized behavior, it fails to resolve the rich internal structure of the critical phase. The SIC, by contrast, reveals a pronounced spatial heterogeneity unique to the critical phase, which we trace back to the incommensurately distributed zeros (IDZs) in the off-diagonal hopping terms. These IDZs partition the chain into nearly disconnected subregions, leading to a stepwise ramp in the steady-state SIC and to coherent long-lived subregion echoes in its dynamics. We further show that these oscillatory echoes are quantitatively explained by a quasiparticle picture in which quasiparticles are confined within the IDZ-delimited subregions, undergoing repeated reflections. To establish the generality of the SIC as a diagnostic tool, we extend our analysis to two additional settings: a phase with a single-particle mobility edge (SPME) separating extended and localized states and a critical phase that does not originate from IDZ fragmentation, and demonstrate that the SIC can cleanly distinguish these scenarios through their distinct steady-state profiles, initial-site sensitivities and the appearance of subregion echoes. Our results establish the SIC as a powerful and readily measurable probe for distinguishing critical phases and for uncovering the bottlenecked connectivity that underlies their multifractal architecture.

The remainder of this paper is organized as follows. Section II introduces the extended Harper model and its phase diagram. Section III presents the half-chain EE dynamics. Section IV is devoted to the SIC, where we discuss the steady-state spatial fingerprints (Sec. IV A) and the subregion echoes arising from IDZ-induced fragmentation with a quasiparticle interpretation (Sec. IV B). In Sec. V, we compare the SIC signatures of the IDZ critical phase with those of two other nontrivial localization scenarios: a SPME phase (Sec. V A) and a critical phase without IDZs (Sec. V B), and summarize their distinguishing features (Sec. V C). We conclude with a summary and outlook in Sec. VI.

II. EXTENDED HARPER MODEL

We consider the 1D extended Harper model [55, 56], whose Hamiltonian is given by

$$H = \sum_j \left[\left(t_j c_j^\dagger c_{j+1} + \text{H.c.} \right) + V \cos(2\pi\beta j + \phi) c_j^\dagger c_j \right], \quad (1)$$

where the nearest-neighbor hopping strength $t_j = 1 + \mu \cos[2\pi\beta(j + \frac{1}{2}) + \phi]$, and c_j ($c_j^\dagger$) denotes the fermionic annihilation (creation) operator at site j . The parameter μ controls the amplitude of the off-diagonal modulation, while V represents the strength of the incommensurate on-site potential. The incommensurate wave vector is chosen as the inverse golden ratio, $\beta = (\sqrt{5} - 1)/2$. For a finite system, it is approximated by $\beta = \lim_{n \rightarrow \infty} (F_{n-1}/F_n)$, where the Fibonacci sequence is defined by $F_n = F_{n-1} + F_{n-2}$ with $F_0 = F_1 = 1$. To impose periodic boundary conditions consistently, we set the system length to $L = F_n$ and take $\beta = F_{n-1}/F_n$. Throughout this work, we fix $\phi = 0$.

The phase diagram is determined by analyzing the fractal dimension. For the m th eigenstate $|\psi_m\rangle = \sum_{j=1}^L u_{m,j} c_j^\dagger |\text{vac}\rangle$, the inverse participation ratio is defined as $\text{IPR}(m) = \sum_j |u_{m,j}|^4$. The corresponding fractal dimension is then obtained from the scaling $\eta = -\lim_{L \rightarrow \infty} \ln \text{IPR} / \ln L$. In the thermodynamic limit, $\eta \rightarrow 1$ ($\eta \rightarrow 0$) indicates an extended (localized) state, and an intermediate value $0 < \eta < 1$ characterizes a critical state. Since the extended Harper model hosts pure phases with no mobility edges, the global phase behavior can be efficiently captured by the mean fractal dimension [28], defined as

$$\bar{\eta} = -\lim_{L \rightarrow \infty} \frac{\ln \text{MIPR}}{\ln L}, \quad (2)$$

where $\text{MIPR} = L^{-1} \sum_{m=1}^L \text{IPR}(m)$ denotes the average inverse participation ratio over all eigenstates. Figure 1(a) displays the resulting phase diagram as a function of μ and V . Region I corresponds to the extended phase ($\bar{\eta} \rightarrow 1$), region II to the critical phase ($0 < \bar{\eta} < 1$), and region III to the localized phase ($\bar{\eta} \rightarrow 0$). In the limit $\mu = 0$, the model reduces to the well-known AA model [15], for which the extended-localized transition occurs at $V = 2$. The Hamiltonian in Eq. (1) exhibits a duality between μ and V under the exchange $\mu \leftrightarrow V/2$ [57]. The self-dual line $V = 2\mu$ corresponds to the critical-to-localized transition, while the extended-to-critical transition at $\mu = 1$ is determined by the pure off-diagonal modulation limit, consistent with the phase diagram shown in Fig. 1(a).

III. ENTANGLEMENT ENTROPY

Following a quantum quench from an unentangled product state, we examine the EE dynamics of the three distinct phases under unitary evolution $|\Psi(t)\rangle = e^{-iHt} |\Psi(0)\rangle$, where H is the Hamiltonian of the extended Harper model specified in Eq. (1). The initial state is chosen as a bipartite configuration $|\Psi(0)\rangle = |11 \dots 100 \dots 0\rangle$, with the left half of the chain fully occupied and the right half empty. The half-chain EE is defined as

$$S(t) = -\text{Tr}[\rho_{L/2}(t) \ln \rho_{L/2}(t)], \quad (3)$$

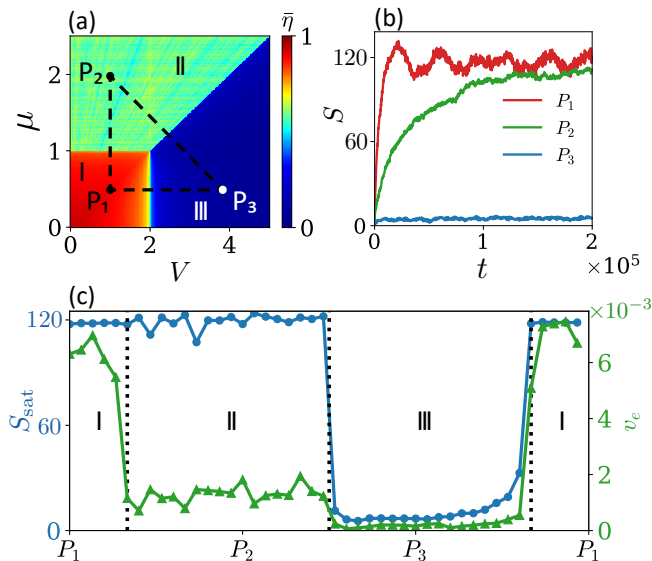


FIG. 1. (a) Phase diagram of the extended Harper model, characterized by mean fractal dimension $\bar{\eta}$ as a function of V and μ for system size $L = F_{14} = 610$. Regions I, II, and III correspond to the extended, critical, and localized phases, respectively. (b) Time evolution of the half-chain entanglement entropy for representative points in the three phases. The parameters (V, μ) are taken from the points labeled P_1 (1.0, 0.5), P_2 (1.0, 2.0), and P_3 (4.0, 0.5) in panel (a). (c) Saturation value S_{sat} (blue curve with dots) and early-time growth velocity v_e (green curve with triangles) as the parameters (V, μ) are varied clockwise along the piecewise path indicated by the dashed line in panel (a): $V = 1.0$, $\mu = 2.5 - 0.5V$, and $\mu = 0.5$, starting from point P_1 . The dotted vertical lines mark the phase boundaries.

where $\rho_{L/2}(t)$ denotes the reduced density matrix of the left half of the system.

Figure 1(b) displays the time evolution of the half-chain EE for representative parameter sets (V, μ) corresponding to the three distinct phases identified in Fig. 1(a): point P_1 (1.0, 0.5) (extended), point P_2 (1.0, 2.0) (critical), and point P_3 (4.0, 0.5) (localized). In the extended phase, the EE grows rapidly and subsequently oscillates around a well-defined saturation value. In the localized phase, the EE saturates at an extremely low value, reflecting strong suppression of entanglement propagation. In the critical phase, the EE eventually approaches a saturation value comparable to that of the extended phase, but the growth is markedly slower, indicating that although entanglement ultimately spreads across the entire system, its propagation is significantly hindered.

To quantitatively distinguish the EE dynamics of different phases, we employ two complementary quantities: the saturation value S_{sat} and the early-time growth velocity v_e [54]. The saturation value S_{sat} is obtained by averaging the EE over a long time interval after it has reached a steady regime; it reflects the system's overall capacity to sustain entanglement. The early-time velocity v_e is

extracted from a linear fit to the initial growth of the EE and characterizes how rapidly entanglement spreads through the system. Figure 1(c) shows S_{sat} and v_e as functions of the parameters (V, μ) varied clockwise along the piecewise path defined by $V = 1.0$, $\mu = 2.5 - 0.5V$ and $\mu = 0.5$ in Fig. 1(a), starting from point P_1 . Regarding the saturation value S_{sat} , the extended and critical phases exhibit similar magnitudes, both substantially larger than that of the localized phase. In contrast, the early-time growth velocity v_e is vastly larger in the extended phase than in the other two phases. A closer comparison reveals that the critical phase possesses a slightly higher v_e than the localized phase. These findings suggest that, from a global perspective, the critical phase resembles the extended phase in allowing entanglement to permeate the entire chain, yet from a microscopic standpoint it behaves more like the localized phase, with pronounced hindrances that slow the initial spread of entanglement.

However, the half-chain EE provides only a coarse-grained view of the distinct phases in a quasiperiodic system, and it is particularly limited in characterizing the critical phase. A complete understanding of the critical phase requires a spatially resolved probe that can track how quantum information navigates the underlying multifractal landscape.

IV. SUBSYSTEM INFORMATION CAPACITY

In this section, we employ the SIC, introduced in Ref. [53], to probe information dynamics from a local, spatially resolved perspective. In contrast to the global view provided by the half-chain EE, the SIC tracks how quantum information, initially localized at a single site, gradually diffuses across or remains trapped within the entire system as a function of the subsystem size. This fine-grained analysis directly exposes the underlying multifractal structure of the critical states.

A. Steady-state SIC

As in the above analysis of the entanglement growth, we consider the same quantum quench protocol starting from the bipartite product state $|11 \dots 100 \dots 0\rangle$. To probe local information dynamics, we introduce a reference qubit R that is maximally entangled with a single site E of the chain, resulting in a Bell pair

$$|\Phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle_E |1\rangle_R - i|1\rangle_E |0\rangle_R), \quad (4)$$

while all other sites remain in an unentangled product state. After the system evolves under the extended Harper Hamiltonian H for a time t , the SIC between the reference qubit R and a subsystem A of length $x = |A|$ centered on site E is defined as the mutual information

$$\text{MI}(x, t) = S_A(t) + S_R(t) - S_{AR}(t). \quad (5)$$

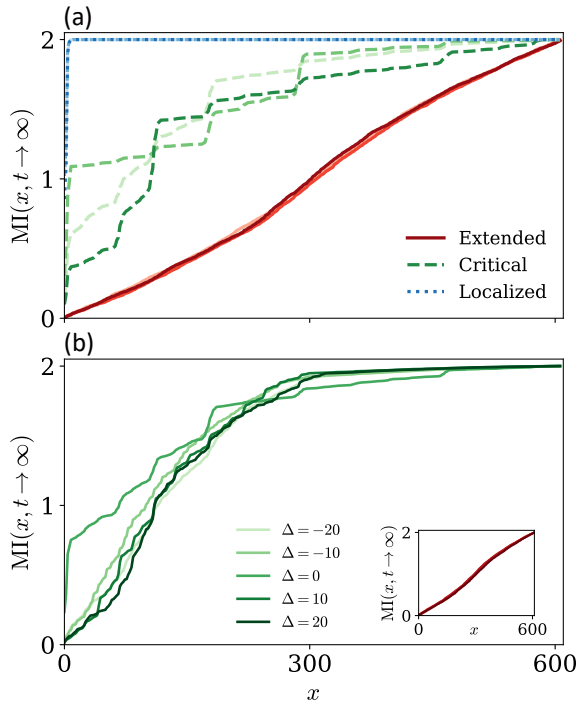


FIG. 2. (a) Steady-state SIC as a function of the subsystem length x following a quantum quench to different phases. The system size is $L = 610$. The red solid lines correspond to the extended phase with parameters $(V, \mu) = (1.0, 0.5)$, $(1.0, 0.6)$, and $(1.0, 0.7)$; the green dashed lines correspond to the critical phase with $(V, \mu) = (1.0, 2.0)$, $(1.2, 1.9)$, and $(1.4, 1.8)$; the blue dotted lines correspond to the localized phase with $(V, \mu) = (4.0, 0.5)$, $(3.8, 0.5)$, and $(3.6, 0.5)$. In each family of curves, the color varies from light to dark in the order listed. (b) Steady-state SIC in the critical phase $[(V, \mu) = (1.0, 2.0)]$ for different spatial offsets Δ of the initially entangled site $E = L/2 + \Delta$. Inset: Steady-state SIC in the extended phase $[(V, \mu) = (1.0, 0.5)]$ for the same set of offsets Δ .

Here, S_α denotes the von Neumann entropy of α ; for example, the entropy of subsystem A is $S_A(t) = -\text{Tr}[\rho_A(t) \log \rho_A(t)]$, with $\rho_A(t)$ the reduced density matrix of A at time t . We adopt the base-2 logarithm so that the mutual information takes values in the interval $MI \in [0, 2]$. The upper bound $MI = 2$ indicates that all of the quantum information initially encoded at site E can be fully recovered from subsystem A .

We first focus on the steady-state SIC, $MI(x, t \rightarrow \infty)$, as a function of the subsystem length x , which captures the long-time spatial distribution of the initially localized information. As shown in Fig. 2(a), the steady-state SIC profile exhibits a linear ramp in the extended phase, reflecting ballistic information transport, and a step-like shape in the localized phase, signaling strong information trapping. These observations are consistent with the findings reported in Ref. [54]. More importantly, our results further demonstrate that the critical phase displays a distinct intermediate behavior: the SIC still

attains its maximum value at $x = L$, confirming that information eventually permeates the entire system, yet its growth with x proceeds via a ramp punctuated by multiple steps, in contrast to the smooth increase observed in the extended phase. This *stepwise* ramp reflects the underlying multifractal structure of the critical states, which imposes spatially sensitive hindrances on the local information propagation.

To further expose this underlying structure, we vary the location of the initially entangled site E by setting $E = L/2 + \Delta$, where Δ denotes a spatial offset from the chain center. We then examine how the steady-state SIC depends on this initial position of the entangled qubit. The results are presented in Fig. 2(b). In the extended and localized phases, introducing a finite offset Δ produces no discernible change in the steady-state SIC profile, as illustrated in the inset for the extended phase. In striking contrast, for the critical phase, different values of Δ induce substantial alterations in the shape of the SIC curve. Notably, for certain offsets, the stepwise ramp characteristic of the critical phase can be largely smeared out or even erased, underscoring the sensitivity of the SIC profile to the exact location of the initially encoded information. This behavior directly indicates that the interior of the critical system is far from homogeneous, unlike the spatially uniform extended and localized phases.

B. Subregion echoes in SIC dynamics

The pronounced spatial inhomogeneity of the critical phase—evidenced by the strong dependence of the SIC on the initial qubit position—suggests that the system is internally fragmented. A natural mechanism for such fragmentation was proposed in Refs. [34, 41], where by employing Avila’s global theory [58], the authors proved that it is the incommensurately distributed zeros (IDZs) in the off-diagonal hopping terms that partition the system into several nearly decoupled segments. We refer to the critical phase generated by this mechanism as the IDZ critical phase. As a result, wave functions in the IDZ critical phase appear delocalized when inspected globally, yet manifest localized features on local scales. This dual character is illustrated in Fig. 3(a), which displays the density distribution of a critical eigenstate of the extended Harper model. The distribution consists of several regions exhibiting strong fractal characteristics, separated by the IDZs (red dots). In the following, we substantiate this picture by exploiting the spatially resolved SIC to directly probe the internal fragmentation of the critical chain.

Following the same quench protocol as in the previous study, we investigate how the SIC evolves inside the subregions delimited by the IDZs. We focus on the two regions labeled B and C in Fig. 3(a). For each region, the initial quantum information is encoded at the central site, and the time-dependent SIC within that region is monitored. Figures 3(b) and 3(d) show the resulting SIC

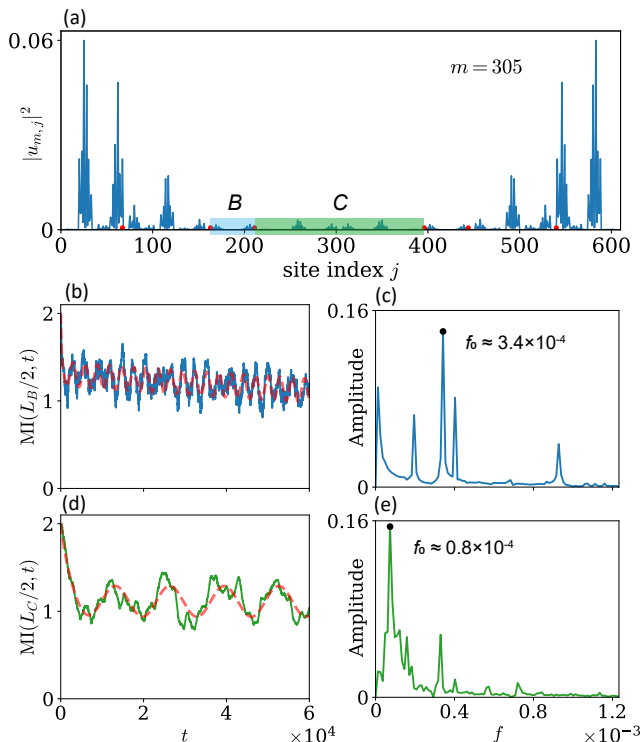


FIG. 3. (a) Spatial probability density of the 305th eigenstate in the critical phase with $(\mu, V) = (2.0, 1.0)$ and $L = 610$. The density exhibits strong fractal characteristics and is partitioned into several segments by the IDZs (red dots), as indicated by the colored regions B and C. (b)–(c) Time evolution of the SIC within region B ($L_B = 48$) and its Fourier spectrum. The initial information is encoded at the center of region B. The subsystem length is set to half of the region. The SIC curve (blue) is fitted by $MI(L_B/2, t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) + (2 - A_1 - A_2) \cos(\omega t)$ (red dashed), where $\omega = 2\pi f_0$ is determined from the dominant peak frequency f_0 in (c). (d)–(e) Corresponding results for region C ($L_C = 185$).

evolutions, which display pronounced oscillatory behavior. To quantify these oscillations, we perform a Fourier spectral analysis, with the corresponding spectra shown in Figs. 3(c) and 3(e). The spectra exhibit clear dominant peaks, revealing oscillation periods of $T_B \approx 3 \times 10^3$ and $T_C \approx 12 \times 10^3$, respectively. Remarkably, the ratio of these periods ($1/4$) closely matches the ratio of the region lengths $L_B/L_C = 48/185$, indicating that the initial information remains largely trapped within each subregion and undergoes long-lived oscillations (echoes) without escaping. We refer to these coherent oscillatory revivals, which arise from quasiparticles repeatedly reflected at the subregion boundaries, as *subregion echoes*.

The observed subregion echoes find a natural explanation within the quasiparticle picture of quantum quench dynamics, a framework originally developed to describe the ballistic growth and saturation of EE in integrable systems [51, 59]. In this picture, the initial product state

serves as a source of entangled quasiparticle pairs that propagate ballistically with opposite momenta. Within the IDZ-delimited subregions, these quasiparticles are largely confined by the hopping bottlenecks: instead of spreading across the entire chain, they undergo repeated reflections at the subregion boundaries, producing coherent echoes of the initially encoded information. The resulting oscillation period is expected to scale as $T \propto L_{\text{region}}/\bar{v}$, where $\bar{v}$ is an effective group velocity of the quasiparticles. Our numerically extracted periods yield a ratio that closely matches the length ratio $T_B/T_C \approx L_B/L_C$, providing strong evidence that the initial quantum information is predominantly carried by quasiparticles trapped within the respective IDZ-delineated subregions. As observed in related studies of finite-size quench dynamics, such oscillatory revivals are a natural consequence of quasiparticles encircling a finite system and returning to their initial location [51, 60], here manifested within each effectively decoupled subregion. The subregion echoes thus provide a dynamical fingerprint of the IDZ-induced bottlenecked connectivity that underpins the multifractal structure of the critical states.

We note that not every subregion separated by the IDZs exhibits clear subregion echoes. In some subregions, the SIC dynamics show irregular or strongly damped oscillations that do not yield well-defined dominant peaks in the Fourier spectrum. The precise conditions under which subregion echoes emerge remain an open question that warrants further investigation. A plausible conjecture is that, although no IDZs are present inside a given subregion, certain sites within the subregion may have anomalously small hopping amplitudes due to the incommensurate modulation of the off-diagonal terms. These local hopping bottlenecks can disrupt ballistic quasiparticle transport, preventing the formation of coherent round-trip reflections and thereby suppressing subregion echoes.

V. COMPARISON WITH OTHER PHASES

The results presented so far establish that the SIC can reveal the internal fragmentation of the IDZ critical phase. A natural question is whether the SIC signatures identified here are specific to the IDZ mechanism, and whether the SIC can distinguish critical phases from other scenarios that also combine extended and localized characteristics. To address these questions, we now apply the same analysis to two contrasting cases: a phase with a SPME, where extended and localized states coexist in the same spectrum, and a critical phase that does not originate from IDZ fragmentation (dubbed non-IDZ critical phase). We compare the SIC signatures of the IDZ critical phase with those of two other nontrivial localization scenarios, demonstrating both the diagnostic power and the generality of the SIC.

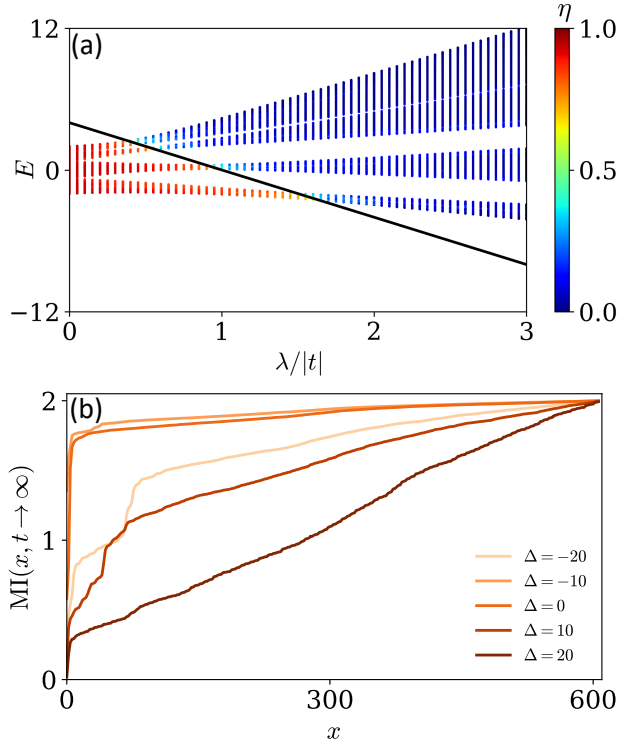


FIG. 4. (a) Energy spectrum of the GAA model for $a = 0.5$ and $L = 610$, with the m th eigenstate colored according to its fractal dimension η . The solid line marks the SPME E_c . (b) Steady-state SIC in the SPME phase at $\lambda = |t|$ for different spatial offsets Δ of the initially entangled site $E = L/2 + \Delta$. The offset dependence manifests mainly in the extent of the early-stage jump.

A. SPME phase

We first examine a SPME phase with coexisting extended and localized states. Although the SPME phase and the critical phase may both exhibit intermediate transport characteristics, their underlying mechanisms are fundamentally different, and we show below that the SIC can cleanly distinguish them.

We consider the generalized Aubry-André (GAA) model [19, 20], described by the Hamiltonian

$$H_{\text{GAA}} = -t \sum_j c_j^\dagger c_{j+1} + \sum_j \epsilon_j c_j^\dagger c_j, \quad (6)$$

where t is the nearest-neighbor hopping strength, and the quasiperiodic on-site potential is given by

$$\epsilon_j = 2\lambda \frac{\cos(2\pi\beta j + \phi)}{1 - a \cos(2\pi\beta j + \phi)}, \quad (7)$$

with λ the potential strength, $\beta = F_{n-1}/F_n$, and the system length $L = F_n$. The parameter a distinguishes this model from the standard AA model; for $a = 0$ the GAA model reduces to the AA model. We set the global phase to $\phi = 0$. The GAA model supports an exact

SPME given by the critical energy [20]

$$aE_c = 2 \operatorname{sgn}(\lambda) (|t| - |\lambda|). \quad (8)$$

In the parameter region where the SPME intersects the energy spectrum, eigenstates with energies above E_c are localized, while those below E_c remain extended. Figure 4(a) displays the energy spectrum of the GAA model, showing clearly the SPME phase situated between a pure extended phase and a pure localized phase.

We compute the steady-state SIC following the same quench protocol as in Sec. IV A, starting from the initial product state $|11\dots 100\dots 0\rangle$ and quenching to the SPME phase, with the reference qubit R initially entangled with site $E = L/2 + \Delta$. As shown in Fig. 4(b), the steady-state SIC exhibits an abrupt initial rise, followed by a steady linear growth, and finally saturates at $x = L$. The initial rise reflects the information trapping effect of the localized states, while the subsequent linear growth indicates ballistic transport mediated by the extended states. The overall profile of the steady-state SIC in the SPME phase differs clearly from that in the pure critical phase studied in Sec. IV A, which displays a ramp punctuated by multiple steps. When the initial encoded site E is varied by the offset Δ , the SIC profile changes mainly in the extent of the early-stage jump, reflecting the different degree of localization in the vicinity of the initial site.

B. Critical phase without IDZs

We next apply the SIC to a non-IDZ critical phase that, in contrast to the extended Harper model, is not generated by explicit IDZs in the off-diagonal hopping. This example tests whether spatial heterogeneity revealed by the SIC is a universal feature of critical phases or is specific to the IDZ mechanism.

We consider a noninteracting 1D spin-orbit (SO) coupled atomic chain subjected to a quasiperiodic on-site potential, as introduced in Ref. [28]. The Hamiltonian reads

$$H_{\text{SO}} = -t_0 \sum_{\langle i,j \rangle} \left(c_{i,\uparrow}^\dagger c_{j,\uparrow} - c_{i,\downarrow}^\dagger c_{j,\downarrow} \right) + \sum_j \delta_j (n_{j,\uparrow} - n_{j,\downarrow}) + \sum_j \left[t_{\text{so}} \left(c_{j,\uparrow}^\dagger c_{j+1,\downarrow} - c_{j,\downarrow}^\dagger c_{j-1,\uparrow} \right) + \text{H.c.} \right]. \quad (9)$$

Here t_0 denotes the spin-conserved nearest-neighbor hopping strength (set to $t_0 = 1$) and t_{so} is the spin-flipped hopping amplitude. The quasiperiodic on-site potential $\delta_j = V \cos(2\pi\beta j + \phi)$ introduces a spin-dependent incommensurate modulation of strength V , with $\beta = F_{n-1}/F_n$ and $L = F_n$. We set the global phase to $\phi = 0$. For $t_{\text{so}} = 0.3$, the model hosts a pure extended phase ($0 < V < 1.4$), a critical phase ($1.4 < V < 2.6$), and a localized phase ($V > 2.6$) [28].

We compute the steady-state SIC after a quench from the initial state $|\uparrow\uparrow\dots\uparrow 00\dots 0\rangle$ to each of these phases,

TABLE I. Comparison of SIC signatures across the SPME phase, the IDZ critical phase, and the non-IDZ critical phase.

	SPME	IDZ critical	Non-IDZ critical
Model	GAA	Extended Harper	SO coupled chain
SIC shape	Abrupt rise + linear ramp	Ramp with steps	Undulating ramp
Initial-site sensitivity	Yes (early rise affected)	Yes	Yes
Subregion echoes	No	Yes	No

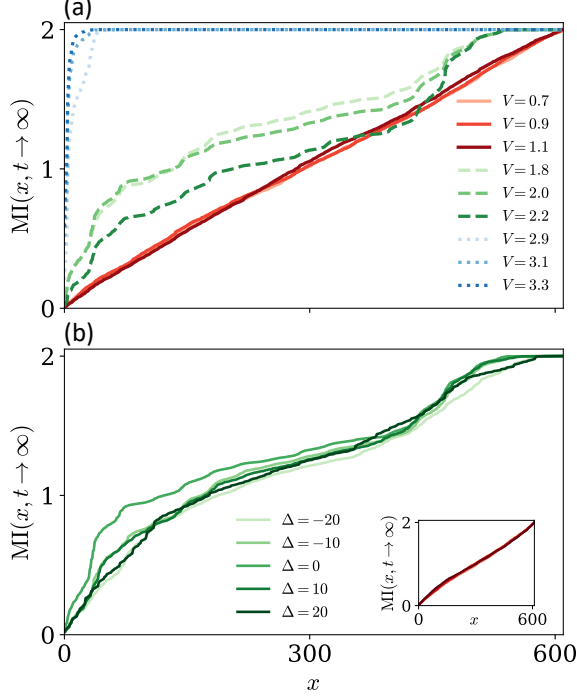


FIG. 5. (a) Steady-state SIC after a quantum quench to different phases in the SO coupled system for $t_{\text{so}} = 0.3$ and $L = 610$. Red solid lines: extended phase ($0 < V < 1.4$); green dashed lines: critical phase ($1.4 < V < 2.6$); blue dotted lines: localized phase ($V > 2.6$). The SIC profile exhibits a clear crossover from a ramp-like shape in the delocalized phase to a step-like shape in the localized phase. (b) Steady-state SIC for different spatial offsets Δ of the initially entangled site in the critical phase (main panel) and in the extended phase (inset). In the extended phase only minor variations are observed under different biases, whereas in the critical phase the biases induce pronounced alterations of the SIC curves.

with the reference qubit R forming a Bell pair with the spin-up state at site $E = L/2$. As shown in Fig. 5(a), the SIC profile displays a clear crossover from a ramp-like shape in the delocalized phase to a step-like shape in the localized phase. The distinction between the critical and extended phases, however, is less pronounced than that found in the extended Harper model: the SIC curve in the critical phase exhibits an undulating profile that grows slightly faster than in the extended phase on average, but lacks the stepwise ramp characteristic of the IDZ

critical phase. This is consistent with the fact that, in contrast to the extended Harper model, this system does not possess explicit IDZs in real space; the critical phase is instead generated by generalized incommensurate zeros in the quasiperiodic on-site potential, as recently proposed for spinful quasiperiodic systems [41]. These zeros can be transformed into IDZs in the hopping coefficients of an effective spinless model. Nevertheless, when the reference qubit R is instead entangled with the spin-up state at a shifted site $E = L/2 + \Delta$, further differences emerge. As shown in Fig. 5(b), varying Δ induces visible alterations of the SIC curve in the critical phase, while the overall undulating shape persists, indicating spatial heterogeneity among different sites of the critical system. In the extended phase, by contrast, only slight variations are observed (see the inset).

C. Summary of SIC signatures

The three localization scenarios investigated in this work—the IDZ critical phase, the SPME phase, and the non-IDZ critical phase—exhibit qualitatively distinct SIC signatures, demonstrating the diagnostic power of this probe. Table I summarizes the key differences.

In the IDZ critical phase, the SIC displays a ramp punctuated by multiple steps in the steady state and pronounced subregion echoes arising from quasiparticle confinement within IDZ-delimited subregions. Both features reflect the internally fragmented structure of the critical chain. The SPME phase, by contrast, produces a steady-state SIC composed of an abrupt initial rise followed by a linear ramp, a direct consequence of the coexistence of localized and extended states at different energies. No oscillatory dynamics is observed, and varying the initial qubit position alters the extent of the early rise. In the non-IDZ critical phase of the SO coupled model, the steady-state SIC exhibits an undulating ramp that grows slightly faster than in the extended phase on average, but lacks the stepwise structure characteristic of the IDZ critical phase. Spatial heterogeneity nevertheless persists: shifting the initial encoding site induces visible alterations in the SIC profile, while its overall undulating shape is preserved, revealing an intrinsic inhomogeneity that is not as pronounced as in the IDZ case. Furthermore, unlike the IDZ critical phase, no regular oscillatory behavior—let alone subregion echoes—is observed in the non-IDZ critical phase. This suggests that the dynamical

fingerprints provided by subregion echoes are a distinctive signature of IDZ-induced fragmentation, rather than a generic feature of all critical phases.

Taken together, these results demonstrate that the SIC can reliably distinguish IDZ critical phases from both SPME phases and non-IDZ critical phases, while also capturing the spatial heterogeneity that appears to be a generic feature of criticality in quasiperiodic systems.

VI. DISCUSSION AND CONCLUSIONS

We have presented a comprehensive study of entanglement dynamics in the extended Harper model, covering its extended, critical, and localized phases. By combining the conventional half-chain EE and the spatially resolved SIC, we have demonstrated that entanglement-based probes can faithfully distinguish the three phases and, more importantly, reveal the internal spatial heterogeneity that sets the critical phase apart. The SIC uncovers a stepwise ramp in the steady state and pronounced oscillatory behavior in the dynamics. Both features originate from the IDZs in the off-diagonal hopping terms, which fragment the chain into weakly connected subregions and confine quasiparticle propagation. The quantitative agreement between the observed oscillation periods and the subregion lengths provides direct dynamical evidence for the IDZ-induced bottleneck picture. By further applying the SIC to two additional scenarios—a phase with a SPME and a critical phase without IDZs—we have demonstrated that the SIC can cleanly distinguish these cases from the IDZ critical phase through their distinct steady-state profiles, initial-site sensitivities, and the presence of subregion echoes, thereby establishing both the diagnostic power and the generality of this probe.

Our findings raise several questions that merit further investigation. First, while the quasiparticle picture successfully accounts for the subregion echoes in the non-interacting limit, it remains to be understood how inter-

particle interactions modify—or possibly wash out—the IDZ-induced fragmentation in interacting generalizations of the extended Harper model. Preliminary studies indicate that genuine many-body critical phases can exist in quasiperiodic systems [61], and the SIC may offer a local window into their intricate entanglement structures. Second, the interplay between the IDZ mechanism and other forms of quasiperiodic modulation, such as those involving long-range hopping [30] or non-Hermitian terms [62, 63], could lead to richer fragmentation patterns and merits systematic exploration. Third, extending the SIC analysis to higher dimensions, where critical phases can occupy finite regions of parameter space [39], is an intriguing direction.

From the experimental perspective, the SIC is particularly appealing because its measurement requires only a single ancilla qubit and a single forward time evolution. Recent advances in the realization of quasiperiodic potentials in cold-atom systems [3–8, 10, 12], photonic lattices [9, 14], and superconducting circuits [11, 13] provide a realistic platform for observing the SIC signatures reported here. Specifically, the steady-state SIC spatial profile can be reconstructed by measuring the mutual information between the reference qubit and a variable-sized subsystem, while subregion echoes can be monitored by tracking time evolution of the subsystem entropies [53]. We expect that such experiments will not only confirm our predictions but also pave the way for using real-space information dynamics as a versatile diagnostic tool for complex quantum phases.

ACKNOWLEDGEMENTS

We acknowledge insightful discussions with Yucheng Wang and Xin-Chi Zhou. This work was supported by the National Natural Science Foundation of China (Grants No. 12204187), and the Innovation Program for Quantum Science and Technology (Grant No. 2021ZD0302000).

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